TECHNICAL WHITEPAPER

Computer Network Topologies

An Architectural Overview of Physical and Logical Network Layouts, Performance Trade-offs, and Design Considerations

PREPARED BY

Prabu Thiagamani

Network Architecture & Engineering Specialist

1. Introduction to Network Topologies

Network Topology defines the structural layout of a telecommunication network, dictating how nodes (computers, routers, switches) are interconnected via communication links. Topology can be viewed in two distinct ways:

- **Physical Topology:** The actual physical arrangement of cables, devices, and hardware locations.
- **Logical Topology:** The architectural path along which data signals travel through the physical network infrastructure.

Selecting the appropriate topology impacts performance, fault tolerance, scalability, and implementation costs. Below is a comprehensive breakdown of the major standard topology models.

2. Core Topology Types

Bus Topology

LINEAR ARCHITECTURE

In a Bus topology, all network nodes are connected sequentially to a single shared central cable, known as the backbone or bus. Data transmitted by any node propagates along the backbone and is available to all connected devices.

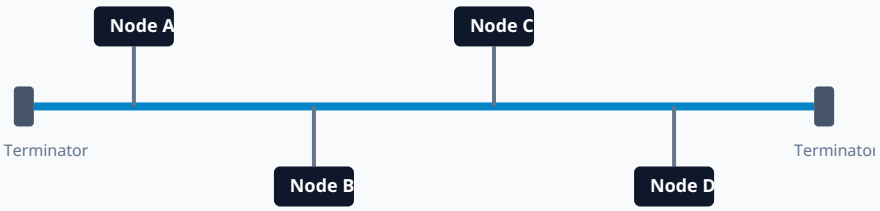

Figure 1: Single linear bus configuration with coaxial cable and termination points.

Advantages (Pros)	Disadvantages (Cons)
• Cost-effective and requires minimal total cabling.	• Single point of failure: cable break disables entire network.
• Easy to set up and expand for small networks.	• High collision rate as network traffic increases.
• No active central device (e.g., switch) required.	• Difficult to isolate and troubleshoot individual faults.

Star Topology

CENTRALIZED POINT-TO-POINT

In a Star topology, all peripheral devices connect to a central hub or switch via point-to-point connections. All network communication travels through the central device, which acts as a signal repeater or packet switcher.

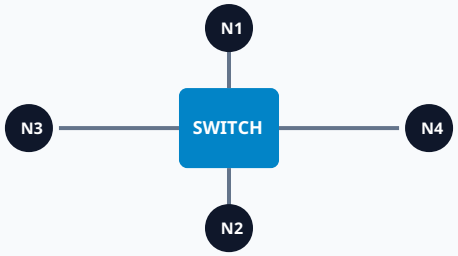

Figure 2: Star layout with central switch handling point-to-point connections.

Advantages (Pros)	Disadvantages (Cons)
• High fault tolerance: single link failure doesn't affect others. • Centralized management and simplified troubleshooting. • Easy to add or remove nodes without network downtime.	• Central node is a single point of failure (switch/hub down = network down). • Higher cable consumption compared to bus topology. • Higher installation costs due to central switch requirements.

Ring Topology

CIRCULAR LOOP

Nodes are connected in a closed continuous loop. Each device is directly connected to two neighboring devices. Data packets travel in one direction (or dual directions in redundant configurations) from node to node through token-passing mechanisms.

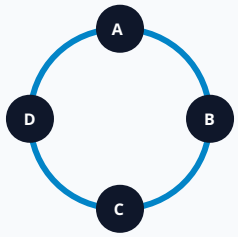

Figure 3: Unidirectional Ring topology demonstrating token propagation.

Advantages (Pros)	Disadvantages (Cons)
• No data packet collisions due to token passing mechanism. • Equal access privileges for all connected computers. • Performs predictably well under heavy network load.	• Unidirectional: single node failure breaks entire network. • Reconfiguration or node addition interrupts communication. • Overall bandwidth limited by the slowest intermediate node.

Mesh Topology

REDUNDANT INTERCONNECTED

In a Full Mesh topology, every node has a direct point-to-point link to every other node in the network. Partial Mesh topologies exist where nodes are connected to multiple, but not all, devices to balance cost and redundancy.

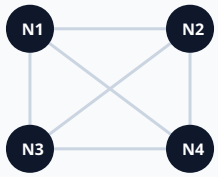

Figure 4: Fully interconnected 4-node Mesh topology. Formula: Links = N(N-1)/2.

Advantages (Pros)	Disadvantages (Cons)
• Maximum redundancy and unmatched fault tolerance. • Dedicated links eliminate channel sharing and traffic congestion. • High security and privacy as data travels on isolated channels.	• Extremely expensive due to massive cabling requirements. • Complex hardware setup and high port density per router/switch. • Difficult installation and maintenance.

3. Architectural Trade-off Analysis

Choosing the ideal topology depends on organizational priorities such as budget, required reliability, and network scale. The comparative table below summarizes key metrics across all topologies:

Topology	Cable Usage	Installation Cost	Fault Tolerance	Troubleshooting	Primary Use Case
Bus	**Very Low**	**Low**	**Very Poor**	**Difficult**	Legacy networks, small setups
Star	**Moderate**	**Moderate**	**High (except Switch)**	**Easy**	Modern LANs, Office networks
Ring	**Low-Moderate**	**Moderate**	**Poor**	**Difficult**	Token Ring, SONET fiber networks
Mesh	**Extremely High**	**Very High**	**Maximum**	**Moderate**	Core Backbones, Data Centers

Topology	Cable Usage	Installation Cost	Fault Tolerance	Troubleshooting	Primary Use Case
Hybrid / Tree	**High**	**High**	**High**	**Structured**	Enterprise WANs, Universities